

Assessment of water quality and of seasonal water use on feedlot and grazing operations in western Kansas operating within the High Plains aquifer

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Abstract

The beef cattle industry is the leading sector of the agriculture economy in Kansas. The objectives of this study were to evaluate the water quality of Kansas grazing and feedyard water samples, as well as quantify feedyard water use by month for 10 feedyards in western Kansas. Sixteen grazing sites in western Kansas were selected to collect forage and water samples. Forage samples from each grazing site were collected in August 2024. Forage clippings from each operation, which together constituted a representative sample of the pasture, were collected and sent to a feed analysis laboratory. Monthly water well meter readings were evaluated for nine of the feedyards as one feedyard did not have accurate historical information. Data were evaluated from January 2020 through June 2025. Water pH ($P < 0.01$) and sulfate ($P = 0.06$) were significantly associated with sampling month for the grazing operations. For the feedyard water quality analyses, water pH ($P < 0.01$), sulfate ($P < 0.01$), phosphorus ($P = 0.01$), calcium ($P < 0.01$), chromium ($P < 0.01$), iron ($P < 0.01$), cobalt ($P < 0.01$), selenium ($P < 0.01$), and molybdenum ($P < 0.01$) were all associated with sampling month; however, magnitude of differences was small by month. Maximum average water use was 14.69 gallons/head/day. Average monthly high temperature was positively associated ($P < 0.01$) and ration dry matter negatively associated ($P = 0.02$) with feedyard water use. These results provide a baseline for benchmarking water conservation efforts.

Key words: beef production, water intake, water conservation, land resource

Introduction

The beef cattle industry is the leading sector of the agriculture economy in Kansas. For the state of Kansas, the counties with the largest annual inventories of cattle are located in western Kansas. The western Kansas counties rely on groundwater resources supplied by the High Plains aquifer. The beef industry in western Kansas is represented by multiple different industry segments from extensive cow-calf/stocker operations grazing native rangelands and irrigated cereal grain pastures to large confined cattle feeding operations. As groundwater

resources are diminished, both the quantity and quality of water will become a primary concern among beef cattle producers in all segments of the industry. Additionally, during periods of peak water demand (i.e. irrigation season, drought) static water levels often decline, which may reduce water quality. Thus, an assessment of aggregate and seasonal water use/demand and water quality in the Kansas beef industry is warranted to characterize the impacts of reduced water availability and water quality, provide baseline information for future benchmarking of conservation efforts, and identifying appropriate mitigation strategies. Providing science based normal cattle drinking water quality ranges is necessary to evaluate potential water filtration systems for reclaiming water to be utilized for potential cattle consumption.

Grazing and feedyard watering systems can be quite extensive. Generally, water is pumped from wells through pipes which are then used to fill the water tanks. Larger feedyards may have a water storage tank which then supply water to the entire feedyard. Multiple wells may be used to supply water in some operations.

Livestock water quality and quantity is rarely studied even though water is arguably the most essential nutrient. To the authors' knowledge, calcium, chromium, iron, cobalt, selenium and molybdenum have never been studied in depth in animal drinking water. Developing normal ranges for these minerals is important to determining what animal drinking water quality is. In most cases normal ranges for minerals in animal drinking water have been based on human drinking water standards.¹ The drinking water standards for animal drinking water should be based on concentrations which may cause adverse health in animals. Future research evaluating water quality samples from multiple feedyards combined with monthly animal health and performance outcomes is warranted. The objectives of this study were to evaluate the water quality of Kansas grazing and feedyard water samples, as well as quantify feedyard water use by month for multiple feedyards in western Kansas. Water quality analysis is critical as is the magnitude of outcome over time for grazing and feedyard operations.

Materials and methods

Forage and water quality evaluation

Sixteen grazing sites in western Kansas were selected to collect forage and water samples. Grazing sites were selected based on agreeing to participate and relationship with investigators. Grazing sites were predominately cow-calf operations utilizing native short-grass pastures and ground water resources from the High Plains aquifer. Predominate grass species on these sites consisted of buffalo grass (*Buchloe dactyloides*), side-oats grama (*Bouteloua curtipendula*), and western wheatgrass (*Pascopyrum smithii*).

Forage samples from each grazing site were collected in August 2024. Forage clippings from each operation, which together constituted a representative sample of the pasture, were collected and sent to a feed analysis laboratory (ServiTech, Dodge City, KS). A consistent protocol was followed across all sites. First, personnel walked a minimum of 10 paces from each watering site (tank). Then, a fixed visual point of reference in the distance was selected, personnel then proceeded an additional 20-50 paces toward the selected point. The sampling area was then selected. A 0.09 m² vegetation sampling hoop^a was then tossed on to the sampling site. A battery-powered shear was then used clip all forage contained within the hoop to a height of approximately 2.54 cm above the ground surface to avoid soil contamination within the sample. Any woody species present within the designated sampling area were not included in the sample and all other standing forage present was clipped and included in the sample for analysis. Thus, all nutrient composition data presented from the sample are the result of both current and any previous year's vegetation from grasses, forbs and grass-like species given the objective was to determine the overall nutrient content of the forage. This protocol was repeated a minimum of 2 times from each watering site, generally proceeding in different directions of travel.

Monthly water samples were collected from hydrants or water tanks (if hydrants could not be accessed) for each grazing site from August through October during the grazing period. Approximately 500 mL of water was collected at each time point in a plastic water sample bottle. Water samples were frozen in a standard deep freezer until shipped to a diagnostic laboratory.

Feedyard water samples were collected monthly from July 2024 to June 2025 from 10 different feedyard operations in western Kansas. One-time capacity for each feedyard included ranged from 30,000 to 90,000 head. Feedyard sites were selected based on agreeing to participate and relationship with investigators. For each month, a single water sample was collected from each feedyard. Feedyard samples were collected from the same water tank within each feedyard. Prior to collecting the sample, the water tank was cleaned and samples were collected directly from where water flowed into the tank. Approximately 500 mL of water was collected at each time point in a plastic water sample bottle. Water samples were frozen in a standard deep freezer until shipped to a diagnostic laboratory.

Water quality analyses for both the grazing and feedyard water samples were performed at Kansas State Veterinary Diagnostic Laboratory (Manhattan, KS). Total dissolved solids (TDS), pH, nitrate, sulfate and mineral concentrations were

analyzed using the laboratory's validated standard operating procedures. The TDS were measured using evaporation, pH was recorded using a pH meter, nitrate and sulfate concentrations were determined via ion chromatography, and mineral concentrations were determined using inductively-coupled plasma mass spectrometry.

Feedyard water use and weather evaluation

Monthly water well meter readings were evaluated for 9 of the feedyards as one feedyard did not have accurate historical information. Data were evaluated from January 2020 through June 2025. Some feedyards were only able to provide accurate historic information for various time points. The number of feedyard operations which were able to provide historic water use and finish ration dry matter for each month is shown in Table 1. Monthly head days from each feedyard were recorded. Monthly water use was divided by head days to determine monthly water use per head.

Data from the closest mesonet weather station^b to each feedyard was extracted each day. Daily high ambient temperatures were averaged for each month from January 2020 to June 2025. The weather data was matched to the feedyard water use data.

Statistical analysis

Descriptive summaries of forage samples from 16 different grazing operations in western Kansas were performed. Data were evaluated using a commercial software program (rStudio Team 2025, Boston, MA). Water quality outcomes were log transformed with a constant of 1 added to outcomes with 0 values (nitrate, phosphorus, chromium, manganese, iron, cobalt, copper, zinc, selenium and molybdenum). Total dissolved solids with values < 1,000 ppm were converted to 1,000 ppm for analyses, as 1,000 ppm was the minimal amount able to be detected in laboratory analysis. Separate analyses were performed for the grazing and feedyard water samples. Models included a fixed effect of sampling month and random effect for repeated measures on location. The difference between high and low values were calculated where sampling month was significant ($P \leq 0.10$) as doing exploratory pilot research observations; otherwise, the average across month was calculated when sampling month was non-significant ($P > 0.10$). Minimum and maximum values for each analyte were reported.

Monthly feedyard water use and average high temperature outcomes were evaluated using linear mixed models with a fixed effect of month and a random effect for repeated measures in individual feedyard. An additional statistical model evaluated the relationship of average high temperature and dry matter on water use with random effect for repeated measures of feedyard. The minimum and maximum values for individual feedyards, along with model-adjusted monthly averages for water use and average high temperature, were summarized.

Results

Forage sample analyses from grazing sites are displayed in Table 2. Water quality analyses by month for grazing (Table 3) and feedyard (Table 4) are shown. Water pH ($P < 0.01$) and sulfate ($P = 0.06$) were significantly associated with sampling month for the grazing operations. For the feedyard water quality analyses, water pH ($P < 0.01$), sulfate ($P < 0.01$), phosphorus ($P = 0.01$), calcium ($P < 0.01$), chromium ($P < 0.01$), iron ($P < 0.01$), cobalt ($P < 0.01$), selenium ($P < 0.01$), and

Table 1: Number of feedyard operations data included for historical water use per head and dry matter by month from nine feedyards in western Kansas.

Year	Month	Water use per head	Finish ration dry matter
2020	January	5	2
	February	5	2
	March	5	2
	April	5	2
	May	5	2
	June	5	2
	July	5	2
	August	5	2
	September	5	2
	October	5	3
	November	5	3
	December	5	3
2021	January	6	4
	February	6	4
	March	6	4
	April	6	4
	May	6	4
	June	6	4
	July	6	4
	August	6	4
	September	6	4
	October	6	4
	November	6	4
	December	6	4
2022	January	6	4
	February	6	4
	March	6	4
	April	6	4
	May	6	4
	June	6	4
	July	6	4
	August	6	4
	September	6	4
	October	6	4
	November	6	4
	December	6	4

Table 1, Cont'd: Number of feedyard operations data included for historical water use per head and dry matter by month from nine feedyards in western Kansas.

Year	Month	Water use per head	Finish ration dry matter
2023	January	6	4
	February	6	5
	March	6	5
	April	6	5
	May	6	5
	June	6	5
	July	8	5
	August	8	5
	September	8	5
	October	8	5
	November	8	5
	December	8	5
2024	January	8	5
	February	8	6
	March	8	6
	April	8	6
	May	8	6
	June	9	6
	July	9	9
	August	9	9
	September	9	9
	October	9	9
	November	9	9
	December	9	9
2025	January	9	9
	February	9	9
	March	9	9
	April	9	9
	May	9	9
	June	9	9

Table 2: Average, standard deviation, minimum and maximum calculated nutrients of grass sample analysis from 16 different grazing sites in western Kansas.

Calculated nutrients	Average	Standard deviation	Minimum	Maximum
Dry matter, %	60.91	13.59	43.00	79.00
Crude protein, %	8.11	1.63	5.20	10.60
Acid detergent fiber, %	43.55	6.61	31.20	50.70
Neutral detergent fiber, %	68.54	7.62	53.40	76.10
Total digestible nutrients, %	53.93	7.50	46.20	67.80
Net energy maintenance, Mcal/lb	0.50	0.12	0.38	0.71
Net energy gain, Mcal/lb	0.25	0.10	0.13	0.44
Calcium, %	0.67	0.38	0.29	1.35
Phosphorous, %	0.15	0.05	0.09	0.25
Magnesium, %	0.19	0.16	0.07	0.50
Potassium, %	1.11	0.96	0.31	2.72
Sulfur, %	0.13	0.05	0.07	0.21
Sodium, %	< 0.01	-	< 0.01	< 0.01
Zinc, mg/kg	24.42	5.25	14.00	32.00
Iron, mg/kg	225.33	112.33	104.00	461.00
Manganese, mg/kg	70.67	55.95	24.00	175.00
Copper, mg/kg	3.58	1.08	2.00	5.00

molybdenum ($P < 0.01$) were all associated with sampling month. Average monthly feedyard water usage and maximum high temperature are shown in Figure 1. Descriptive summary of model-adjusted and individual feedyard water use and average high temperatures from nine feedyards in western Kansas are shown in Table 5. Average monthly high temperature was significantly ($P < 0.01$) associated with feedyard water use as well as ration dry matter ($P = 0.02$; Figure 2).

Discussion

While more water quality outcomes changed over time for the feedyard water samples compared to the grazing site water samples, it is important to note the feedyard samples were collected over a year whereas the grazing water samples were collected over a 3-month period consistent with a grazing season in western Kansas. Total dissolved solids were higher in the grazing site water samples compared to the feedyard water samples. The grazing site water samples were collected from water sources (hydrants) or established water-tanks (if source was not accessible), but water tanks were not cleaned prior to sampling. Whereas, feedyard water samples were collected from a clean tank to remove feed particles and sediment which may impact analyses. Samples were obtained in a manner to minimize contaminants and sediment within the tanks. Due to the nature and construction of the tanks utilized from grazing sites (large concrete, tire or permanent galvanized ring tanks) it would have been impractical to drain and clean watering sites used at the ranch sites. Although tanks from the grazing site operations were not cleaned prior to sampling, water samples were obtained directly from or in close proximity to the water supply for the site (hydrant or pipe).

Feedyard water samples were collected from the same water tank each month to represent the water source to which the cattle had access. Feedyards may have multiple wells supplying water. The feedyard results may have been different if samples were collected at individual wells, but we wanted to collect samples which were representative of the entire feedyard.

High levels of total dissolved solids and sulfate have reduced average daily gain, dry matter intake, and water intake in growing beef steers; however, in the report steers were consuming 4,800 ppm total dissolved solids and 3,000 ppm sulfate.² Water samples in the current study are much lower for total dissolved solids and sulfate. Total dietary sulfur in cattle diets includes the drinking water and the diet. The National Research Council recommends a maximum tolerable level of 0.3% total sulfur for low forage diets including sulfur in the drinking water and sulfur in the diet.¹ Safe maximum concentration of sulfate in drinking water for dairy heifers was reported to be 2,500 ppm.^{3,4} The current consensus recommendation is that water sulfate should not exceed 500 ppm and 1,000 ppm for calves and adult cows, respectively. High sulfur content in drinking water is an issue throughout the United States.⁵ Depending on dietary sulfur concentration, water with 1,000 to 1,500 ppm of sulfate may cause antagonism of copper, selenium, manganese and zinc.⁶⁻⁸

The 5 criteria most often considered in assessing water quality for both humans and livestock are organoleptic properties (odor and taste); physiochemical properties (pH, total dissolved solids, total dissolved oxygen, and hardness); presence of toxic compounds (heavy metals, toxic minerals, organophosphates, and hydrocarbons); presence of excess minerals or compounds (nitrates, sodium, sulfates and iron);

Table 3: Model-adjusted least square means ($\pm$ SE) of water quality analysis by month in 16 different grazing sites in western Kansas. Models included random effect of repeated measures for individual site.

Water quality	Month			SE	P value	Difference [¶]	Average [†]	Minimum [‡]	Maximum [#]
	August	September	October						
Total dissolved solids, ppm	1,235.54	1,247.21	1,238.48	1.11	0.89		1,240.41	1,000.00	3,200.00
pH	7.84	8.22	8.11	1.01	<0.01	0.38		6.81	9.94
Nitrate, ppm	9.56	10.04	9.06	0.29	0.80		9.55	2.35	90.40
Sulfate, ppm	15.35	15.59	16.64	1.48	0.06	1.28		1.57	176.97
Sodium, ppm	53.13	56.29	54.66	1.38	0.19		54.69	7.50	553.56
Magnesium, ppm	27.07	25.70	24.32	1.34	0.16		25.70	3.23	166.73
Phosphorous, ppm	0.06	0.07	0.05	0.02	0.35		0.06	0.01	0.63
Potassium, ppm	5.95	6.11	5.78	1.15	0.21		5.95	2.46	24.28
Calcium, ppm	72.49	65.70	70.98	1.22	0.49		69.73	13.64	284.54
Chromium, ppm	0.000	0.001	0.001	0.000	0.45		0.001	0.000	0.002
Manganese, ppm	0.028	0.029	0.010	0.020	0.58		0.023	0.000	0.477
Iron, ppm	0.193	0.176	0.202	0.074	0.95		0.190	0.001	1.725
Cobalt, ppm	0.000	0.000	0.000	0.000	0.14		0.000	0.000	0.002
Copper, ppm	0.011	0.006	0.043	0.015	0.17		0.020	0.000	0.470
Zinc, ppm	0.160	0.064	0.159	0.054	0.16		0.128	0.001	1.725
Selenium, ppm	0.009	0.008	0.008	0.004	0.43		0.008	0.000	0.064
Molybdenum, ppm	0.006	0.005	0.006	0.001	0.19		0.004	0.000	0.024

^{||} Largest standard error in analysis

[¶] Difference between high and low value where significant ($P \leq 0.10$)

[†] Average across months where non-significant ($P > 0.10$)

[‡] Minimum value observed in raw data

[#] Maximum value observed in raw data

Table 4: Model-adjusted least square means ($\pm$ SE) of water quality analysis by month in 10 different feedyard operations in western Kansas beginning in July 2024 and concluding June 2025. Models included random effect of repeated measures for individual site.

Water quality	Month							
	July	August	September	October	November	December	January	February
Total dissolved solids, ppm	1,000.20	1,000.00	999.76	1,002.19	1,002.19	1,000.00	1,000.00	1,000.00
pH	8.35	8.02	8.05	8.07	8.02	8.12	8.20	7.85
Nitrate, ppm	16.59	15.55	11.03	18.65	19.38	17.78	20.76	18.82
Sulfate, ppm	55.16	61.17	6.51	14.18	17.25	15.54	20.83	32.01
Sodium, ppm	43.00	43.81	58.14	42.35	43.24	41.09	43.49	42.24
Magnesium, ppm	17.33	17.59	18.46	18.97	19.06	16.92	18.10	17.25
Phosphorous, ppm	0.23	0.00	0.27	0.05	0.03	0.02	0.07	0.02
Potassium, ppm	4.59	4.56	4.89	4.67	4.67	4.48	4.68	4.40
Calcium, ppm	40.75	43.15	44.33	56.49	58.83	50.09	59.34	68.44
Chromium, ppm	0.006	0.009	0.008	0.001	0.001	0.001	0.001	0.004
Manganese, ppm	0.003	0.005	0.011	0.003	0.000	0.018	0.087	0.007
Iron, ppm	0.206	0.006	0.001	0.012	0.006	0.021	0.068	0.040
Cobalt, ppm	0.001	0.001	0.001	0.000	0.000	0.000	0.000	0.000
Copper, ppm	0.001	0.001	0.001	0.004	0.034	0.001	0.007	0.001
Zinc, ppm	0.022	0.011	0.010	0.028	0.024	0.015	0.028	0.019
Selenium, ppm	0.014	0.013	0.009	0.003	0.004	0.004	0.004	0.004
Molybdenum, ppm	0.008	0.007	0.008	0.006	0.006	0.006	0.005	0.005

Table 4, Cont'd: Model-adjusted least square means ($\pm$ SE) of water quality analysis by month in 10 different feedyard operations in western Kansas beginning in July 2024 and concluding June 2025. Models included random effect of repeated measures for individual site.

Water quality	Month				SE	P value	Difference [¶]	Average [†]	Minimum [‡]	Maximum [#]
	March	April	May	June						
Total dissolved solids, ppm	1,018.40	1,018.40	1,000.00	999.74	1.01	0.38		1,003.41	1,000.00	2,000.00
pH	7.87	7.88	7.68	7.62	1.01	< 0.01	0.73		7.13	9.29
Nitrate, ppm	17.81	15.76	17.55	18.00	0.22	0.54		17.31	0.00	55.21
Sulfate, ppm	42.60	24.65	27.80	37.99	1.56	< 0.01	54.66		1.78	640.22
Sodium, ppm	42.32	42.50	40.31	42.44	1.15	0.26		43.74	23.42	562.16
Magnesium, ppm	18.42	18.25	18.42	18.05	1.23	0.89		18.07	6.42	70.96
Phosphorous, ppm	0.01	0.03	0.03	0.09	0.06	0.01	0.27		0.00	2.37
Potassium, ppm	4.54	4.37	4.46	4.95	1.10	0.80		4.61	2.04	17.75
Calcium, ppm	74.88	70.52	70.21	79.99	1.17	< 0.01	39.24		21.99	254.45
Chromium, ppm	0.001	0.001	0.001	0.001	0.001	< 0.01	0.008		0.000	0.030
Manganese, ppm	0.002	0.003	0.002	0.011	0.028	0.39		0.013	0.000	1.262
Iron, ppm	0.006	0.037	0.013	0.044	0.040	< 0.01	0.205		0.000	2.120
Cobalt, ppm	0.000	0.000	0.000	0.001	0.000	< 0.01	0.001		0.000	0.002
Copper, ppm	0.000	0.001	0.001	0.001	0.009	0.21		0.004	0.000	0.341
Zinc, ppm	0.026	0.031	0.009	0.045	0.016	0.74		0.022	0.000	0.354
Selenium, ppm	0.004	0.003	0.004	0.004	0.002	< 0.01	0.011		0.001	0.033
Molybdenum, ppm	0.005	0.005	0.005	0.004	0.001	< 0.01	0.003		0.000	0.013

^{||} Largest standard error in analysis

[¶] Difference between high and low value where significant ($P \leq 0.10$)

[†] Average across months where non-significant ($P > 0.10$)

[‡] Minimum value observed in raw data

[#] Maximum value observed in raw data

Figure 1: Model adjusted (95% confidence interval) monthly feedyard water use (blue line) and average high temperature (red line) from nine feedyards in western Kansas. Statistical models included fixed effect for each month and random effect for repeated measures in individual feedyard.

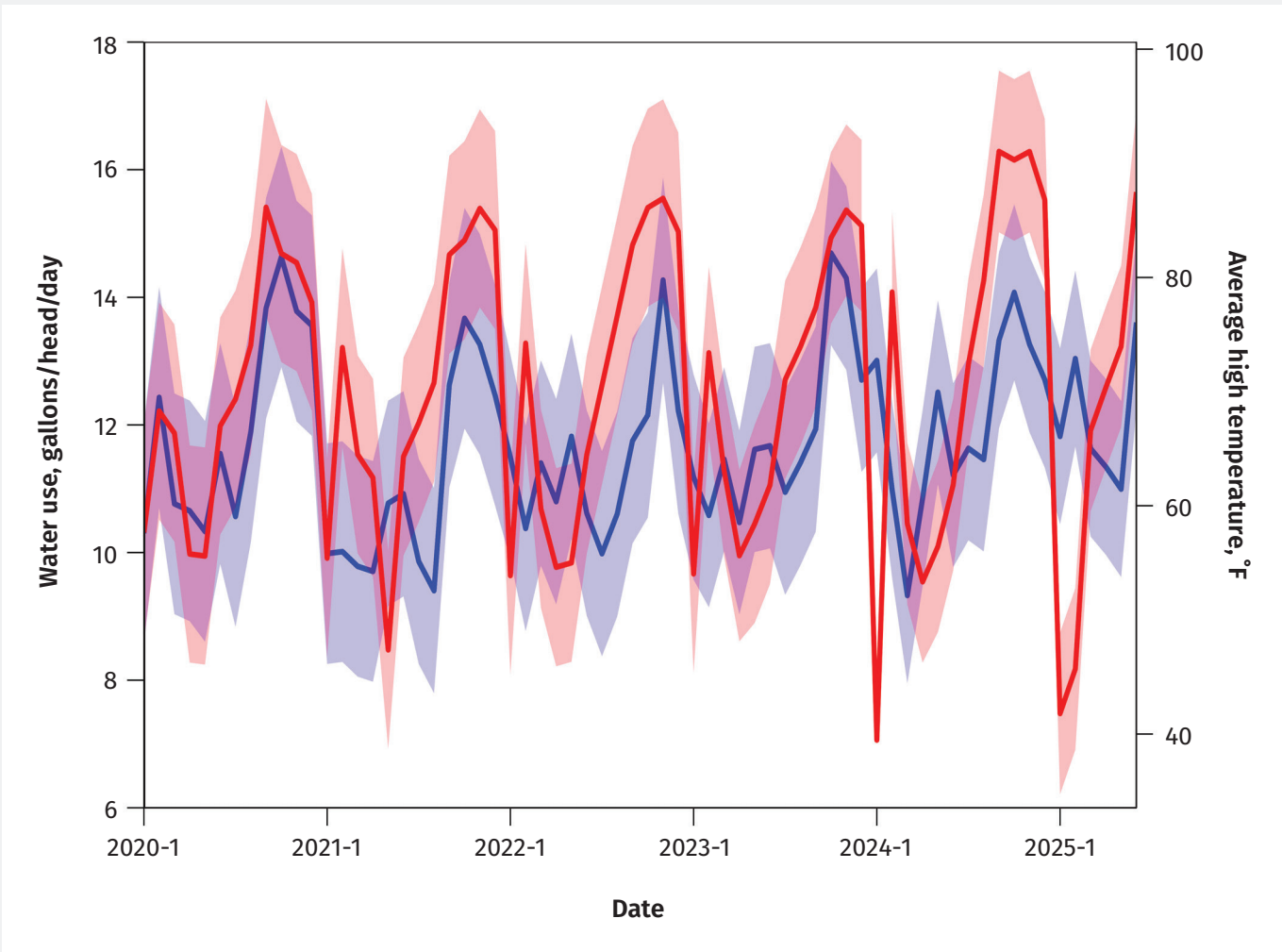
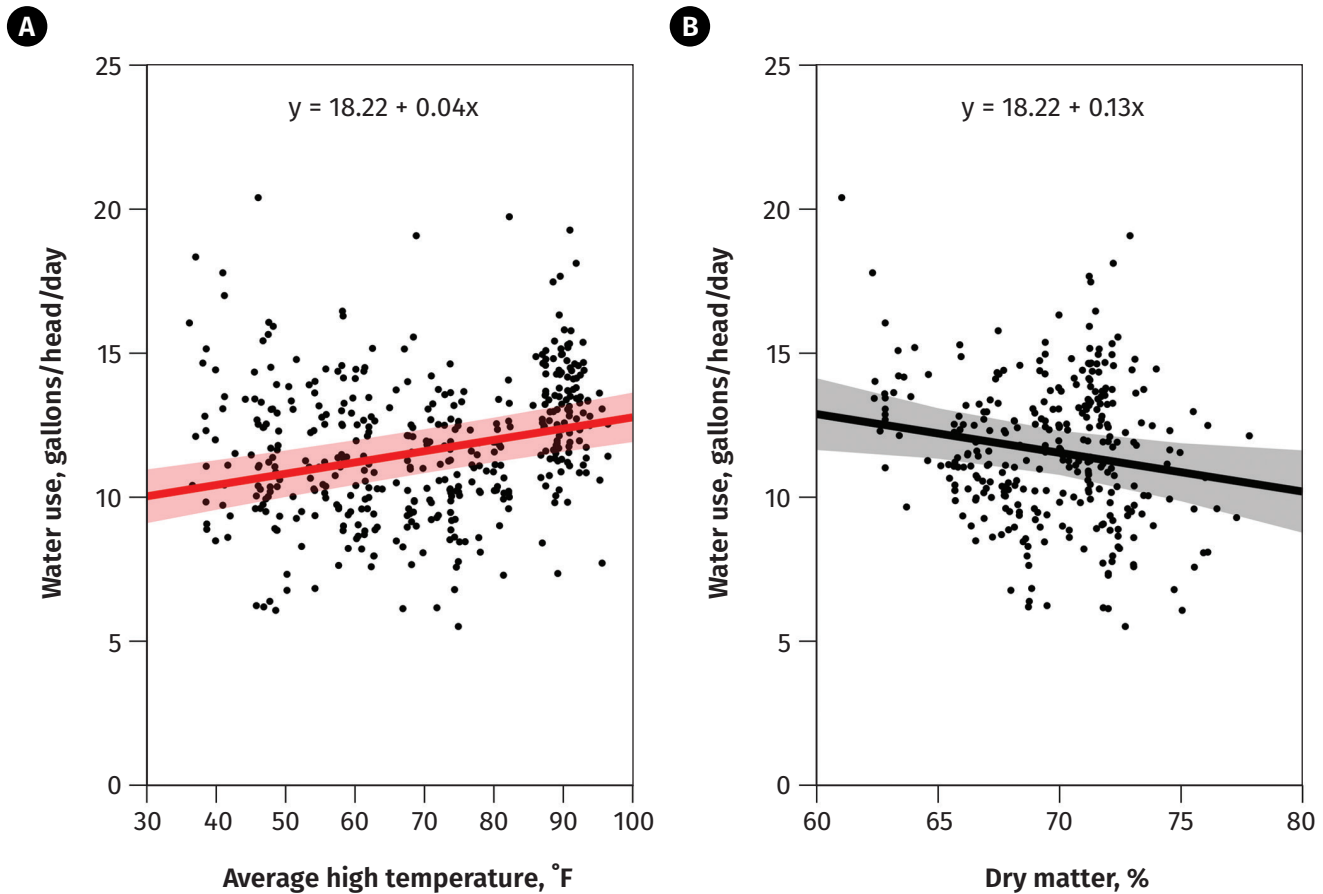


Table 5: Descriptive summary of model-adjusted and individual feedyard water use and average high temperatures from nine feedyards in western Kansas.

Summary	Model-adjusted water use, gallons/head/day	Individual feedyard water use, gallons/head/day	Model-adjusted average monthly high temperature, °F	Individual feedyard average monthly high temperature, °F
Minimum	9.32	5.51	39.47	36.14
Quartile 1	10.68	10.17	58.74	57.14
Median	11.52	11.73	70.66	71.19
Mean	11.72	11.76	70.00	70.10
Quartile 3	12.69	13.27	82.07	87.48
Maximum	14.69	20.40	91.04	96.43

Figure 2: Feedyard water use regression by average monthly high temperature and finish ration dry matter from nine feedyards in western Kansas. Model included fixed effects of average monthly high temperature, ration dry matter, and random effect for repeated measures of feedyard. Average monthly high temperature ($P < 0.01$) and ration dry matter ($P = 0.02$) was a significant predictor of water use.



and presence of bacteria.¹ There has been no peer-reviewed work on odor and taste in livestock drinking water. There is very little to no data on the effect on livestock drinking water concerning odor and taste. The United States Environmental Protection Agency provides guidelines for human consumption, but recommendations for livestock consumption based on data are not available.^{9,10} To the authors' knowledge, this is the first study to quantify the water quality in grazing and feedyard operations over a period of time. Larger sample sizes and data over a longer period of time are needed to adequately evaluate animal health and performance outcomes related to water quality.

The pH of water is typically neutral to alkaline. Feedyard cattle routinely have bouts of subclinical acidosis while consuming a highly digestible ration and utilize consumption of water to help balance rumen pH.¹¹ Calcium in water is a measure of water hardness which interestingly increased with additional rainfall (data not shown) in the area during the second half of the sampling period. During the same period, the water pH decreased. While phosphorous, chromium, iron, cobalt, selenium, and molybdenum statistically changed throughout the year, it is important to differentiate between statistical and biological significance.¹² The magnitude of difference between these outcomes is relatively small. Overall, it appears a single water sample is sufficient to evaluate water quality for cattle

based on the magnitude of change throughout the grazing season or year for feedyard; however, if an outcome is close to excessive outcomes, it is warranted to perform multiple tests as well as monitor health and performance outcomes.

Maximum average water use was 14.69 gallons/head/day which is slightly lower than the 16 gallons/head/day previously described in the feedyard.¹³ The peak water use corresponded with higher ambient temperatures which agrees with previous projects.¹⁴ Growing beef cattle of 608 lbs consumed 7.9 gallons/head/day.¹⁵ An additional study evaluated individual animal water intake on crossbred steers with a range of 8.52 to 13.24 gallons/head/day with animals that consumed more water had higher dry matter intakes and higher average daily gain.¹⁶ Average monthly cattle body weight across the feedyard and dry matter intake information was not available for the current study. Another study evaluated daily water intake of 4.57 gallons/head/day during the winter months and 8.55 gallons/head/day during the summer months in feedlot cattle in Nebraska.¹⁷ While the current study has overall greater water use compared to previous studies, water use would have included all activities on the feedyard including mill use, cleaning and water consumption. Knowledge of the water use requirements for feedyards is important considering the additional days on feed and larger finish weights targeted in current production systems.

Evaluation of the relationship between average high temperature, dry matter, and water use provided some surprising results. As expected, there was a positive relationship with average high temperature and water use. Surprisingly, water use was lower with higher finish ration dry matter. The higher finish ration dry matter has less moisture included in the ration, so the authors hypothesized cattle would consume more water to meet the requirements. Finish ration dry matter was chosen as it represents the greatest proportion of ration fed to cattle while at the feedyard, but starter and beta agonist rations are commonly fed as well. As stated previously, monthly average body weight and dry matter intake information was not available as data are not commonly recorded in the feedyard industry. Additional research is needed to further evaluate the relationship of water use and finish ration dry matter.

Conclusion

These results are necessary for evaluating grazing and feedyard water quality in western Kansas. Water quality by month is important to consider the magnitude of outcome over time for grazing and feedyard operations. These results provide a baseline for benchmarking water conservation efforts.

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Conflicts of interest

The authors declare there is no conflict of interest for this project.

Use of artificial intelligence (AI) assisted technologies

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Author contributions

All authors were involved in conception and study design. All authors were involved with manuscript drafting and/or revising process. All authors have approved the final version of this manuscript for publication.

Endnotes

^a Forestry Suppliers, Jackson, Mississippi

^b Kansas Mesonet, 2025: Kansas Mesonet Historical Data. Accessed 10 November 2025, <http://mesonet.k-state.edu/weather/historical>

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